Chapter 10 — Privacy, Security, Trust Domains, Treaties, and the Constitutional Firewall

Purpose and scope

Defines information classification, privacy purpose, secrets, credentials, trust boundaries, external domains, Treaties, and governed exchange.

Source authority

Book I: Book I privacy, dignity, sovereignty, external trust, and protected authority
Book II: Book II Chapters 18, 19, 20, 21, 25, and 26
Book III: Book III Chapters 02, 04, 05, 06, and 07

Normative semantic rules

• External-domain exchange MUST be modeled through an External Trust Domain, an applicable Treaty, and the Constitutional Firewall.

• Every active Treaty MUST be exact, scoped, time-bounded, revocable, auditable, and explicitly Owner-authorized through the Owner Authorization Ceremony bound to that exact Treaty.

• A Treaty MUST NOT grant Authority prohibited by Book I or bypass constitutional enforcement.

• Data use MUST state classification, authorized purpose, minimization, access, retention, disclosure, and disposal rules.

• Security Controls and Authority Controls MUST be distinguished even when implemented by the same mechanism.

Canonical terms

HAL-TERM-0116 — External Trust Domain
An external environment, organization, system, or authority regime whose controls and assumptions are not natively governed by HAL.
Distinction: External status does not imply hostility or trustworthiness; exchange requires explicit governance.
Type/category/status: External governance domain / Trust / Approved
Aliases: ETD
Source basis: Direct source normalization
Book I: Book I Decision 49
Book II: Book II Chapters 20 and 21
Book III: Book III Chapter 5
Example: A dependent artifact cites `HAL-TERM-0116` when it uses **External Trust Domain** with this exact governed meaning: An external environment, organization, system, or authority regime whose controls and assumptions are not natively governed by HAL.
Counterexample: A dependent artifact uses **External Trust Domain** in a way that violates its required distinction: External status does not imply hostility or trustworthiness; exchange requires explicit governance.
Relationships: HAL-REL-0041, HAL-REL-0050
Lifecycle: None registered
Constraint: External status does not imply hostility or trustworthiness; exchange requires explicit governance.

HAL-TERM-0117 — Treaty
An exact, scoped, time-bounded, revocable, auditable, and explicitly Owner-authorized agreement defining allowed cross-domain identities, data, capabilities, authority, obligations, verification, monitoring, failure behavior, and termination.
Distinction: Activation requires the Owner Authorization Ceremony bound to the exact Treaty. Conversation, delegated ordinary authority, trust, usefulness, prior access, or the Treaty itself cannot grant authority prohibited by Book I or bypass the Constitutional Firewall.
Type/category/status: Governed trust agreement / Trust / Approved
Aliases: External Trust Treaty
Source basis: Direct source normalization
Book I: Book I Decision 49
Book II: Book II Chapter 21
Book III: Book III Chapters 5 and 7

Example: A dependent artifact cites `HAL-TERM-0117` when it uses **Treaty** with this exact governed meaning: An exact, scoped, time-bounded, revocable, auditable, and explicitly Owner-authorized agreement defining allowed cross-domain identities, data, capabilities, authority, obligations, verification, monitoring, failure behavior, and termination.
Counterexample: A dependent artifact uses **Treaty** in a way that violates its required distinction: Activation requires the Owner Authorization Ceremony bound to the exact Treaty. Conversation, delegated ordinary authority, trust, usefulness, prior access, or the Treaty itself cannot grant authority prohibited by Book I or bypass the Constitutional Firewall.
Relationships: HAL-REL-0041, HAL-REL-0042
Lifecycle: HAL-TRANS-0018, HAL-TRANS-0019, HAL-TRANS-0020
Constraint: Activation requires the Owner Authorization Ceremony bound to the exact Treaty. Conversation, delegated ordinary authority, trust, usefulness, prior access, or the Treaty itself cannot grant authority prohibited by Book I or bypass the Constitutional Firewall.

HAL-TERM-0118 — Constitutional Firewall
The Book II boundary that mediates and enforces constitutionally governed exchange between HAL and External Trust Domains.
Distinction: It is not merely a network firewall and must not be bypassed by direct integration.
Type/category/status: Architectural enforcement boundary / Trust / Approved
Aliases: CF
Source basis: Direct source normalization
Book I: Book I Decision 49
Book II: Book II Chapter 20
Book III: Book III Chapter 5
Example: A dependent artifact cites `HAL-TERM-0118` when it uses **Constitutional Firewall** with this exact governed meaning: The Book II boundary that mediates and enforces constitutionally governed exchange between HAL and External Trust Domains.
Counterexample: A dependent artifact uses **Constitutional Firewall** in a way that violates its required distinction: It is not merely a network firewall and must not be bypassed by direct integration.
Relationships: HAL-REL-0042
Lifecycle: None registered
Constraint: It is not merely a network firewall and must not be bypassed by direct integration.

HAL-TERM-0119 — Trust Boundary
A boundary across which identity, authority, integrity, confidentiality, provenance, or governance assumptions change.
Distinction: Network adjacency alone does not define or erase a Trust Boundary.
Type/category/status: Security boundary / Security / Approved
Aliases: None
Source basis: Direct source normalization
Book I: Book I Decisions 26, 27, 32, 37, 39, 48, and 49
Book II: Book II Chapters 18, 19, 20, 21, 25, and 26
Book III: Book III Chapters 2, 4, 5, 6, and 7
Example: A dependent artifact cites `HAL-TERM-0119` when it uses **Trust Boundary** with this exact governed meaning: A boundary across which identity, authority, integrity, confidentiality, provenance, or governance assumptions change.
Counterexample: A dependent artifact uses **Trust Boundary** in a way that violates its required distinction: Network adjacency alone does not define or erase a Trust Boundary.
Relationships: None registered
Lifecycle: None registered
Constraint: Network adjacency alone does not define or erase a Trust Boundary.

HAL-TERM-0120 — Data Classification
A governed label determining handling, access, transmission, retention, evidence, and disposal requirements for information.
Distinction: Classification is not a purpose or permission to use the data.
Type/category/status: Governance label / Privacy / Approved
Aliases: None
Source basis: Direct source normalization
Book I: Book I Decision 32
Book II: Book II Chapter 19
Book III: Book III Chapters 4 and 5
Example: A dependent artifact cites `HAL-TERM-0120` when it uses **Data Classification** with this exact governed meaning: A governed label determining handling, access, transmission, retention, evidence, and disposal requirements

for information.
Counterexample: A dependent artifact uses **Data Classification** in a way that violates its required distinction: Classification is not a purpose or permission to use the data.
Relationships: HAL-REL-0043
Lifecycle: None registered
Constraint: Classification is not a purpose or permission to use the data.

HAL-TERM-0121 — Personal Data
Information relating to an identified or reasonably identifiable human under the governing privacy context.
Distinction: Pseudonymization may reduce exposure but does not necessarily remove personal-data status.
Type/category/status: Information class / Privacy / Approved
Aliases: None
Source basis: Direct source normalization
Book I: Book I Decision 32
Book II: Book II Chapter 19
Book III: Book III Chapters 4 and 5
Example: A dependent artifact cites `HAL-TERM-0121` when it uses **Personal Data** with this exact governed meaning: Information relating to an identified or reasonably identifiable human under the governing privacy context.
Counterexample: A dependent artifact uses **Personal Data** in a way that violates its required distinction: Pseudonymization may reduce exposure but does not necessarily remove personal-data status.
Relationships: None registered
Lifecycle: None registered
Constraint: Pseudonymization may reduce exposure but does not necessarily remove personal-data status.

HAL-TERM-0122 — Sensitive Data
Information whose unauthorized use or disclosure could materially harm a person, HAL, the Owner, security, sovereignty, trust, or protected operations.
Distinction: Sensitivity is context-dependent and may include non-personal operational data.
Type/category/status: Information class / Privacy / Approved
Aliases: None
Source basis: Direct source normalization
Book I: Book I Decision 32
Book II: Book II Chapter 19
Book III: Book III Chapters 4 and 5
Example: A dependent artifact cites `HAL-TERM-0122` when it uses **Sensitive Data** with this exact governed meaning: Information whose unauthorized use or disclosure could materially harm a person, HAL, the Owner, security, sovereignty, trust, or protected operations.
Counterexample: A dependent artifact uses **Sensitive Data** in a way that violates its required distinction: Sensitivity is context-dependent and may include non-personal operational data.
Relationships: None registered
Lifecycle: None registered
Constraint: Sensitivity is context-dependent and may include non-personal operational data.

HAL-TERM-0123 — Purpose Limitation
The rule that governed data may be collected, used, shared, and retained only for declared, authorized, compatible purposes.
Distinction: Availability or technical usefulness does not establish purpose.
Type/category/status: Use constraint / Privacy / Approved
Aliases: None
Source basis: Direct source normalization
Book I: Book I Decision 32
Book II: Book II Chapter 19
Book III: Book III Chapters 4 and 5
Example: A dependent artifact cites `HAL-TERM-0123` when it uses **Purpose Limitation** with this exact governed meaning: The rule that governed data may be collected, used, shared, and retained only for declared, authorized, compatible purposes.
Counterexample: A dependent artifact uses **Purpose Limitation** in a way that violates its required distinction: Availability or technical usefulness does not establish purpose.
Relationships: None registered
Lifecycle: None registered
Constraint: Availability or technical usefulness does not establish purpose.

HAL-TERM-0124 — Data Minimization
The rule that only the least data reasonably necessary for an authorized purpose may be collected, processed, retained, and disclosed.
Distinction: Minimization applies to fields, precision, population, duration, access, and derived inferences.
Type/category/status: Collection and use constraint / Privacy / Approved
Aliases: None
Source basis: Direct source normalization
Book I: Book I Decision 32
Book II: Book II Chapter 19
Book III: Book III Chapters 4 and 5
Example: A dependent artifact cites `HAL-TERM-0124` when it uses **Data Minimization** with this exact governed meaning: The rule that only the least data reasonably necessary for an authorized purpose may be collected, processed, retained, and disclosed.
Counterexample: A dependent artifact uses **Data Minimization** in a way that violates its required distinction: Minimization applies to fields, precision, population, duration, access, and derived inferences.
Relationships: None registered
Lifecycle: None registered
Constraint: Minimization applies to fields, precision, population, duration, access, and derived inferences.

HAL-TERM-0125 — Retention Class
A governed category defining retention period, review, legal or constitutional hold behavior, archival, and disposal requirements.
Distinction: A Retention Class does not itself authorize collection or access.
Type/category/status: Lifecycle label / Privacy / Approved
Aliases: None
Source basis: Direct source normalization
Book I: Book I Decision 32
Book II: Book II Chapter 19
Book III: Book III Chapters 4 and 5
Example: A dependent artifact cites `HAL-TERM-0125` when it uses **Retention Class** with this exact governed meaning: A governed category defining retention period, review, legal or constitutional hold behavior, archival, and disposal requirements.
Counterexample: A dependent artifact uses **Retention Class** in a way that violates its required distinction: A Retention Class does not itself authorize collection or access.
Relationships: HAL-REL-0043
Lifecycle: None registered
Constraint: A Retention Class does not itself authorize collection or access.

HAL-TERM-0126 — Secret
Confidential material whose disclosure could enable unauthorized access, impersonation, decryption, signing, or protected operation.
Distinction: An Identifier or public key is not a Secret merely because it is security-related.
Type/category/status: Sensitive authentication material / Security / Approved
Aliases: None
Source basis: Direct source normalization
Book I: Book I Decisions 27, 37, and 39
Book II: Book II Chapters 05, 19, and 26
Book III: Book III Chapters 2 and 5
Example: A dependent artifact cites `HAL-TERM-0126` when it uses **Secret** with this exact governed meaning: Confidential material whose disclosure could enable unauthorized access, impersonation, decryption, signing, or protected operation.
Counterexample: A dependent artifact uses **Secret** in a way that violates its required distinction: An Identifier or public key is not a Secret merely because it is security-related.
Relationships: None registered
Lifecycle: None registered
Constraint: An Identifier or public key is not a Secret merely because it is security-related.

HAL-TERM-0127 — Cryptographic Key
A governed value used by an approved cryptographic operation and managed through generation, storage, access, rotation, revocation, destruction, and provenance controls.
Distinction: A key's possession does not itself establish business Authority.
Type/category/status: Cryptographic material / Security / Approved

Aliases: None
Source basis: Direct source normalization
Book I: Book I Decisions 27, 37, and 39
Book II: Book II Chapters 05, 19, and 26
Book III: Book III Chapters 2 and 5
Example: A dependent artifact cites `HAL-TERM-0127` when it uses **Cryptographic Key** with this exact governed meaning: A governed value used by an approved cryptographic operation and managed through generation, storage, access, rotation, revocation, destruction, and provenance controls.
Counterexample: A dependent artifact uses **Cryptographic Key** in a way that violates its required distinction: A key's possession does not itself establish business Authority.
Relationships: None registered
Lifecycle: None registered
Constraint: A key's possession does not itself establish business Authority.

HAL-TERM-0128 — Security Control
A safeguard intended to protect confidentiality, integrity, availability, identity assurance, or system resilience.
Distinction: A Security Control protecting HAL is distinct from an Authority Control preventing HAL from exceeding its mandate.
Type/category/status: Protective control / Security / Approved
Aliases: None
Source basis: Direct source normalization
Book I: Book I Decisions 27, 37, and 39
Book II: Book II Chapters 05, 19, and 26
Book III: Book III Chapters 2 and 5
Example: A dependent artifact cites `HAL-TERM-0128` when it uses **Security Control** with this exact governed meaning: A safeguard intended to protect confidentiality, integrity, availability, identity assurance, or system resilience.
Counterexample: A dependent artifact uses **Security Control** in a way that violates its required distinction: A Security Control protecting HAL is distinct from an Authority Control preventing HAL from exceeding its mandate.
Relationships: None registered
Lifecycle: None registered
Constraint: A Security Control protecting HAL is distinct from an Authority Control preventing HAL from exceeding its mandate.

HAL-TERM-0129 — Authority Control
A safeguard that prevents a Principal, component, or HAL from deciding or acting beyond governed Authority.
Distinction: It is not interchangeable with a Security Control, though one mechanism may support both.
Type/category/status: Mandate-limiting control / Authority / Approved
Aliases: None
Source basis: Direct source normalization
Book I: Book I Decisions 27, 37, and 39
Book II: Book II Chapters 05, 19, and 26
Book III: Book III Chapters 2 and 5
Example: A dependent artifact cites `HAL-TERM-0129` when it uses **Authority Control** with this exact governed meaning: A safeguard that prevents a Principal, component, or HAL from deciding or acting beyond governed Authority.
Counterexample: A dependent artifact uses **Authority Control** in a way that violates its required distinction: It is not interchangeable with a Security Control, though one mechanism may support both.
Relationships: None registered
Lifecycle: None registered
Constraint: It is not interchangeable with a Security Control, though one mechanism may support both.

HAL-TERM-0163 — Trust Domain
A bounded identity, policy, evidence, security, privacy, and accountability context within which trust assumptions and controls are evaluated.
Distinction: Trust Domain is the generic concept. An External Trust Domain is a Trust Domain outside HAL's native governance boundary and requires governed cross-domain exchange.
Type/category/status: Governance context / Trust / Approved
Aliases: None
Source basis: Derived semantic synthesis required by the Book X scope
Book I: Book I Decisions 49 and 52
Book II: Book II Chapters 18, 21, and 24
Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0163` when it uses **Trust Domain** with this exact governed meaning: A bounded identity, policy, evidence, security, privacy, and accountability context within which trust assumptions and controls are evaluated.
Counterexample: A dependent artifact uses **Trust Domain** in a way that violates its required distinction: Trust Domain is the generic concept. An External Trust Domain is a Trust Domain outside HAL's native governance boundary and requires governed cross-domain exchange.
Relationships: HAL-REL-0050
Lifecycle: None registered
Constraint: Trust Domain is the generic concept. An External Trust Domain is a Trust Domain outside HAL's native governance boundary and requires governed cross-domain exchange.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship register. Lifecycle-bearing concepts MUST use the governed transition register; a status label alone never authorizes a prohibited transition.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit.

Anti-patterns

Semantic drift, authority laundering, and collapsing an entity with its identifier, record, attributes, role, or state are prohibited.

Verification

Verify ID and label uniqueness, precise source traceability, source-basis classification, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-format content parity.

Change and deprecation

Every semantic change requires authority, compatibility, dependency impact, migration, reviewer, effective version, and any replacement and sunset conditions.

Review findings

Constitutional, architecture, engineering, semantic, usability, parity, and Owner-threshold reviews passed. No Owner Review item.